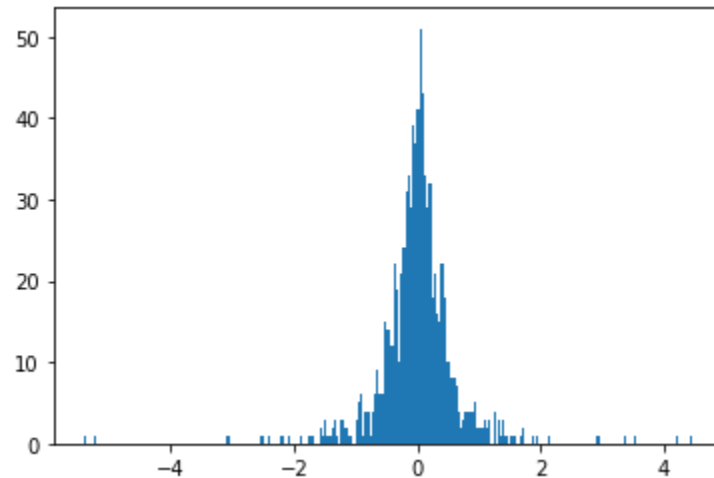


SC4172 Tiny Machine Machine Learning Tutorial 2 – Quantizations

The figure below shows the histogram of the Weights of a trained model, with values range between minimum of -5.4 and maximum of +4.5 (in float32 data type)



A quantization scheme is to be used to reduce(compress) the size of model for storage.

1. What are the desirable properties of the quantization scheme?
2. Discuss how we can (theoretically) reduce/compress the file size by up to 4 times using quantization.

Decrease size,
minimal loss
accuracy etc.

3. Propose an encoding scheme to achieve this. State the encoding and decoding formula used in the proposed quantization scheme, and calculate the maximum error introduced by the scheme during conversion.

Changing float of 32 bits to 8 bit integer, from 4 bytes to 1 byte. It looks at the total range and place the values in range of 0 - 255

4. In addition to compress the size of the data for storing, it is proposed that computation during inference is to be done in the compressed domain (i.e., using the encoded values directly). Would this be possible based on the proposed scheme, and if not, what could be the alternative(s)?
5. Activation functions in the model will also be involved in the computation of the values during the inference operation. Discuss whether it is possible to use the same encoding approach (used for the Weights) to quantize the activation values? If not, what can be done to achieve this.

4. Quantised code = $\text{uint8}(((\text{original} - \text{min}) / (\text{max} - \text{min})) * 255)$, reconstructed = $((\text{quantised_code} / 255.0) * (\text{max} - \text{min})) + \text{min}$, snap to lower grid point of the quantisation unit, typically flooring the value, if rounding to nearest int is used, then snaps to nearest grid point. Max error = $\pm (\text{range} / 256) / 2$ where range = max-min, assumption is that it snaps to nearest int instead of flooring. To perform calculation directly using encoded int it is impt to represent:

- both +ve and -ve values
- maintain zero value exactly, impt as 0 appears more than other numbers

adjustment to the scheme, use absolute value maximum of the min/max range of the values, from -5.4 to +5.4, exclude -128 in the encoded domain such that we have equal +ve and -ve value, linear relationship across whole range.

5. activation function is not known until input data is applied as such it is not possible to apply quantisation to activation function. But we can do post training calibration to estimate the output data range expected of an activation function by running the model using training/ test set data and measure the output value then apply the encoding scheme that can cover the maximum range observed at the output.

- what is there are outliers?
- clip the output to range specified.